

Semantic Monitoring for Adaptive Compute Allocation in Edge Robotic Vision-Language Systems

*A thesis submitted in partial fulfillment of the requirements
for the award of the degree of MASTER OF TECHNOLOGY in INFORMATION TECHNOLOGY*



with specialization in HUMAN COMPUTER INTERACTION

Submitted by

DEVARAKONDA SHANMUKHA SAI
(MHC2024013) *Under the Supervision of*

PROF. G. C. NANDI *to the*

DEPARTMENT OF INFORMATION TECHNOLOGY भारतीय सूचना प्रौद्योगिकी संस्थान, इलाहाबाद

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, ALLAHABAD June 2026

SEMANTIC MONITORING FOR ADAPTIVE COMPUTE ALLOCATION IN EDGE ROBOTIC VISION-LANGUAGE SYSTEMS
M.TECH • IITA • 2026

DEVARAKONDA SHANMUKHA SAI